

RESEARCH ARTICLE

Short-term response of fish communities and water chemistry to breaching of a causeway in the Sarita River Estuary, British Columbia, Canada

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On the west coast of Vancouver Island, Canada, in the Treaty Settlement Lands of the Huu-ay-aht First Nation, a causeway isolating the southern portion of the Sarita River estuary was breached in 2018 to enhance juvenile salmon and tidal water access into the southern portion of the estuary. Short-term goals of breaching were to: (1) enhance juvenile salmon access through the causeway; (2) promote colonization of Chinook and Chum Salmon in pools, while retaining Coho Salmon; (3) monitor the entire estuary fish community, including non-salmonids, to assess breaching success; and (4) homogenize water chemistry in pool and channel ecosystems currently fragmented by the causeway. Immediately following the breach, both fish community and water chemistry measurements indicate that these goals were achieved. Chinook and Chum Salmon utilization increased inside and outside of the causeway following breaching. Similarly, Chinook and Chum salmon were observed in pool ecosystems, and Coho Salmon were also retained in pools following breaching. Water chemistry was homogenized post-breach, primarily due to increased saltwater penetration. While continued monitoring is required, short-term indicators suggest that breaching was effective in increasing connectivity of the Sarita Estuary, allowing access to 1.4 km² of habitat that had previously been inaccessible to juvenile salmon.

Key words: dike, dyke, pipefish, remediation, restoration, Salmon, Sculpin

Implications for Practice

- Breaching had immediate impacts upon water chemistry and fish communities.
- Water chemistry was homogenized around the causeway, primarily due to increased saltwater intrusion.
- Juvenile salmon usage increased around the causeway.
- Breaching may be an effective tool to reduce the fragmenting impact of estuary causeways.

Introduction

Located where rivers discharge freshwater into the ocean, estuaries represent dynamic and species-rich coastal habitats (Simenstad et al. 1982; Elliott & Whitfield 2011). Not only are estuaries biodiversity hotspots for plants, birds, fish, and invertebrates (Lotze et al. 2006; Elliott et al. 2007), but estuaries are also important nursery grounds for Pacific salmon (Simenstad et al. 1982; Moore et al. 2016; Kennedy et al. 2018). Estuaries likewise offer critical environmental services, such as flood reduction and water filtration (Zedler 2000; Elliott et al. 2007; Elliott & Whitfield 2011). Therefore, beyond their intrinsic value, estuaries are economically, culturally, and nutritionally important to humans.

Globally, estuaries are heavily impacted by human activities, as over 40% of the world's population lives within 100 km of

the coast (Small & Nicholls 2003). Large cities located on or upstream of estuaries result in contamination of coastal ecosystems by human and industrial effluents (Howarth 1998; Elliott et al. 2007). Further, as they grant access to inland habitats, estuaries are often hubs of industrial activities such as coastal logging (Simenstad et al. 1982; Musick et al. 2000), as well as the site of metal smelters or pulp and paper mills (Yunker et al. 2011; Gerwing et al. 2018). However, one of the most obvious forms of human alteration in estuaries are dikes and causeways, built to enhance transportation through estuaries, as well as to convert wetlands/marshlands into habitat more suitable for development or farming (Eertman et al. 2002; Hood 2004). These sources of degradation can have negative impacts not only upon estuaries but also upon human populations

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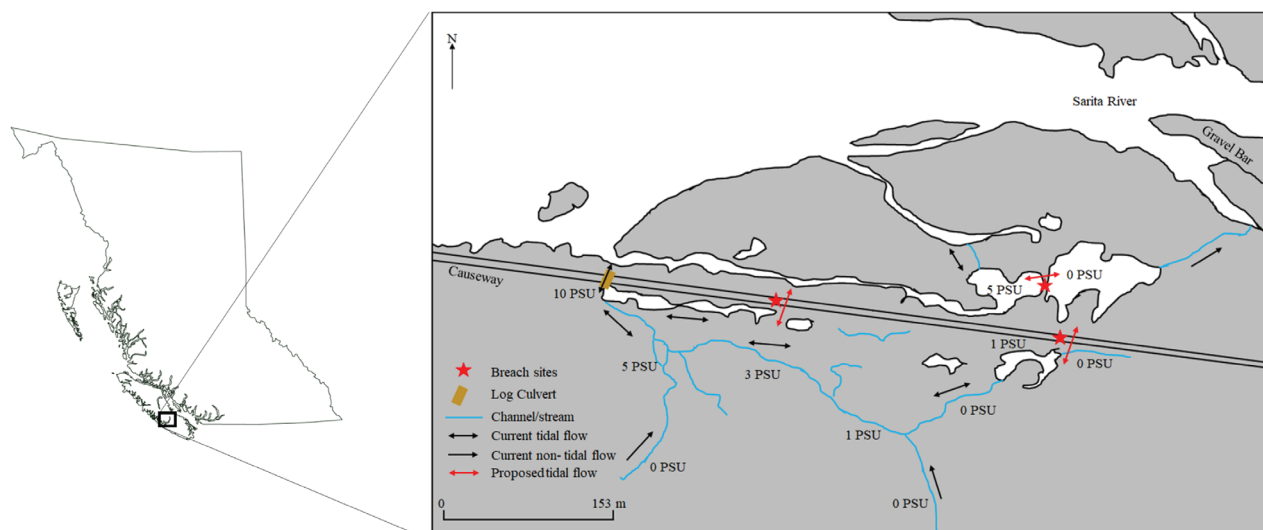


Figure 1. Location of the Sarita River causeway, breach sites, as well as pre-breach and post-breach flow directions. PSU values represent practical salinity units and were measured at the surface during low tide pre-breaching.

that rely upon them. As such, considerable resources have been invested in estuary restoration (Zedler 2000; Herbold et al. 2014).

While restoration of estuaries must address all degrading factors, stressors are often addressed individually due to financial and logistical constraints (Elliott et al. 2007; Peterson & Lowe 2009; Borja et al. 2010). Common targets are dikes and causeways, given the obvious nature of their impact, coupled with the equally obvious solution: breaching or removing the obstruction (Gray et al. 2002; Roni et al. 2002; Gerwing et al. 2017). Dikes and causeways fragment estuary habitat, restricting movement of water, nutrients, sediment, animals, and plant propagules (Bray et al. 1982; Roni et al. 2002; Hood 2004; van Proosdij et al. 2009). Removing the barrier, or installing culverts/bridges, can have immediate positive impacts not only upon hydrology and sedimentation within the estuary but also upon plant and animal communities (McAlice & Jaeger 1983;

Shreffler et al. 1990; Gray et al. 2002; Hood 2004; Morand & Haralampides 2006).

Located along the west coast of Vancouver Island, British Columbia, Canada, the traditional territories and Treaty Settlement Lands of the Maa-nulth First Nations (Huu-ay-aht First Nations, Ka:'yu:'k't'h'/Che:k'tles7et'h First Nations, Toquaht Nation, Uchucklesaht Tribe, and Ucluelet First Nation) encompass numerous estuaries that support commercially and culturally important salmon fisheries. Of particular interest is the estuary of the Sarita River (Figs. 1 & 2), located on the Treaty Settlement Lands of the Huu-ay-aht First Nations. The Sarita River and its estuary support important salmon fisheries; however, the productivity of these fisheries is currently depressed. Further, the Sarita watershed has been impacted by anthropogenic activities, including logging, and the construction of a causeway in the southern portion of the estuary (Ochman 2014). The causeway was built between 1946 and

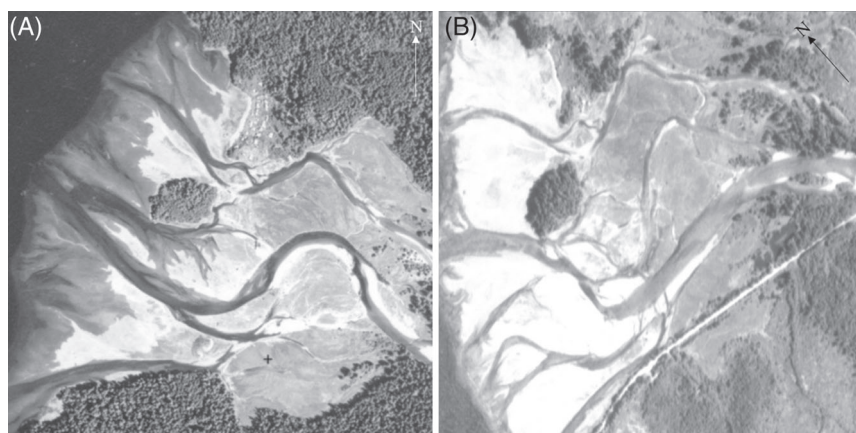


Figure 2. Aerial images of the Sarita River estuary in 1927 (A) and 1976 (B) showing the impact the causeway had upon wetland habitat in the southern portion of the Sarita River estuary.

1954 along the southeast side of the estuary (Figs. 1 & 2), allowing for the construction of a road providing access to the Sarita logging camp. Two large pools excavated north of the eventual causeway were used to acquire construction material. After cessation of cattle ranching following World War II, the estuary was left to proceed through passive vegetative succession; however, the road was maintained.

Currently, tidal water and salmon are restricted in accessing wetland habitat south of the causeway and do so through a single log culvert (Figs. 1 & 2). North of the causeway, the pool furthest east is usually cut off from tidal inputs, except during peak tides; therefore, at low tide most of the pool is freshwater (0–1 PSU, practical salinity units) with a thin layer of saline water near the bottom (3 m; 5 PSU). Small numbers of overwintering Coho Salmon (*Oncorhynchus kisutch*) have been observed in this pool. Conversely, the western pool receives consistent tidal inputs; consequently, the salinity of this pool is higher at all depths (approximately 5 PSU at the surface during low tide). Channels containing eelgrass (*Zostera marina*) beds have developed on either side of the causeway, and the pools north of the causeway also support eelgrass beds. Salinity is much higher in channels outside of the causeway than inside (Fig. 1). Such water chemistry heterogeneity over limited spatiotemporal scales highlights the level of fragmentation imposed upon this landscape by the causeway. Further, expansive mussel beds (*Mytilus* spp.) were observed under the existing log culvert. While the causeway has limited penetration of tidal water and salmon into the wetlands south of the causeway, channels and pools may represent high-quality rearing habitat for juvenile salmon, given overhanging banks and vegetation, as well as eelgrass beds (Simenstad et al. 1982; Carr-Harris et al. 2015; Kennedy et al. 2018). Unfortunately, local knowledge holders report that salmon, both juveniles and adults, used the southern portions of the estuary extensively before causeway construction, however, annual utilization declined drastically after construction (personal communication).

To restore salmon access to potentially high-quality rearing habitat south of the causeway, the causeway was breached in 2018. Breaching of the causeway was predicated upon the following short-term objectives:

- 1 Enhance juvenile salmon access through the causeway to the southern portion of the Sarita Estuary.
- 2 Promote colonization and rearing of Chinook (*Oncorhynchus tshawytscha*) and Chum Salmon (*Oncorhynchus keta*) in the pools, while also retaining Coho Salmon.
- 3 Monitor estuary fish community, including non-salmonids, to assess breaching success.
- 4 Homogenize water chemistry between pools.
- 5 Homogenize channel water chemistry conditions on both sides of the causeway.

Long term, it is hoped that increased access to high-quality estuary rearing habitat will contribute to stabilization and/or enhancement of salmon stocks in the Sarita watershed. While the causeway in the Sarita Estuary represents only one stressor among many (such as logging and climate change), estuaries and the nursery grounds they provide are critical habitat in

the life history of Pacific salmon (Simenstad et al. 1982; Carr-Harris et al. 2015; Kennedy et al. 2018). Restoration efforts upstream in the watershed may fail to improve salmon production if sufficient estuary habitat is not available to support increasing populations of salmon as they forage amongst eelgrass beds and acclimatize to more saline waters. Restoration efforts within estuaries may therefore remove a potential bottleneck with regards to salmon survival.

Methods

The road over the causeway was deactivated in 2018, and the causeway was breached in two locations, the log culvert was removed, and a third breach made between the pools (Fig. 1). Breaches were completed between 14 and 27 March, 2018, and rip rap was added to stabilize banks and limit release of silt and clay into the water. The deactivated road was seeded with local grasses and seedlings in spring 2019 to promote vegetative regeneration.

Data Collection

The Sarita Estuary was divided into three sampling areas: sites near the causeway (Inside: channels south of the causeway and Outside: channels north of the causeway), pools (Freshwater and Saltwater), and areas removed from the causeway but still in the estuary (Estuary: North Beach (NB), South Beach (SB), and Island (Isl); Fig. 3). Estuary sites were used as reference locations, areas that would not be impacted by breaching. South Beach is a gravel/cobble beach, with small patches of eelgrass located subtidally. Over the past 2 years, 2017/2018, this site was observed to have the highest and most consistent captures of juvenile salmon in the estuary (Unpublished data; $n = 26–30$; SB: 0.36 ± 0.34 salmon/m²; NB: 0.0008 ± 0.0008 salmon/m²; Island: 0.08 ± 0.08 salmon/m²). North Beach is also dominated by gravel and cobble; however, substantial eelgrass beds are located subtidally. Finally, the Island site is located in a deep pool east of the Island denoted in Figures 1–3. Eelgrass is abundant in this pool, and the shore is dominated by fine sediments, saltmarsh, and braided channels. When observed together, these reference locations represent the range of habitat available in the Sarita Estuary for juvenile salmon.

Goals 1–3: Enhancement of Salmonid Access and Monitoring of the Estuary Fish Community.

To determine if breaching increased salmon usage of pool or channel habitats, and how the fish community responded to breaching, beach seining was conducted throughout the estuary. Beach seining targeting juvenile salmonids and other estuary fish species was conducted at all locations before and after the causeway was breached (February to April 2018; 1–2 days pre-breach; 7 days post-breach). Exact seining dates can be found in Table S1. Sampling occurred between February and April as this is the period when juvenile Chinook and Chum Salmon are migrating down river and residing within the estuary. A net 10 m long, 1.8 m high, and with mesh openings of 3.18 mm was used for all seines. At all

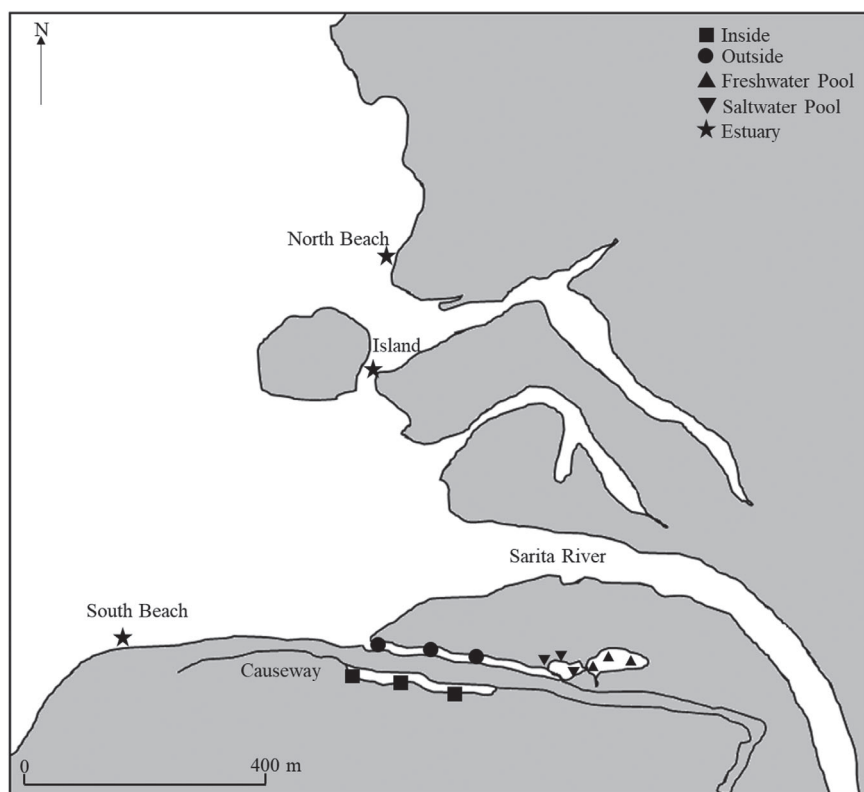


Figure 3. Map of beach seining locations in the Sarita Estuary. Locations inside and outside the causeway, as well as those in the Saltwater and Freshwater Pools indicate a single seine. At North and South Beach, as well as the Island site, three replicate seines separated by 10 m were conducted.

estuary sites, three replicate seines, separated by 10 m, were conducted. Seining locations inside and outside the causeway, as well as in the pools were singular (Figs. 2 & 3). However, all seines were replicated in time (inside and outside causeway, as well as Saltwater and Freshwater Pools: pre-breach $n = 2$, post-breach $n = 7$; Island, North and South Beach: pre-breach $n = 1$, post-breach $n = 5$). For each seine, the net was pulled on one end in a perpendicular direction away from shore until a maximum depth of approximately 1.2 m was reached. The net was pulled back to shore in a sweeping motion while trying to cover a minimum area of 100 m². Seines were conducted on a falling tide.

Goals 4 and 5: Homogenization of Water Chemistry in Channels and Pools. To assess if breaching homogenized water chemistry in pools and channels, water chemistry was monitored pre- and post-breaching, just prior to beach seining, using a YSI 2030 (dissolved oxygen [DO], specific conductivity, and temperature, from which salinity was calculated; APHA 2012) and a Hanna HI98129 pH pen. At seining locations in channels inside and outside of the causeway, water chemistry was only quantified at the surface. In pools, profiles were collected with readings taken every meter; however, pH was only measured at the surface. Each profile was measured in triplicate and from the center of each pool at high tide.

Statistical Analyses

Statistical analyses were conducted in a stepwise manner, with two categories of response variables: fish community and water chemistry. With regards to the fish community, we first performed a permutational multivariate analysis of variance (PERMANOVA; Anderson et al. 2008) to quantify if breaching impacted the observed fish community as a whole. As part of the PERMANOVA, a priori planned contrasts were used to determine which locations varied before and after breaching. Locations that varied significantly were further examined using a similarities percentages (SIMPER) analysis to determine how individual species, including salmonids, varied between locations. Water chemistry values were assessed also using a PERMANOVA to determine if the aquatic environment changes as a whole before or after breaching.

Goals 1–3: Enhancement of Salmonid Access and Monitoring of the Estuary Fish Community. In the PERMANOVA, a Before-After-Control-Impact (BACI) design was utilized (Underwood 1994, 1997) to contrast fish communities in pool and causeway sites before and after the breach, as well as with reference sites. The response variable for the PERMANOVA was a resemblance matrix assessing community similarities between seines, with similarity calculated using Bray-Curtis coefficients, and fish community data were

fourth-root-transformed to improve assessment of effects of rare and common taxa on community structure (Clarke 1993). In the PERMANOVA random factors included area (three levels: causeway, pools, and estuary), site nested within area (Site [Area]); seven levels: Inside and Outside causeway, Freshwater and Saltwater Pools, North and South Beach, and Island), and date nested in breach (Date [Breach]; 8 levels, see Table S1 for dates). Breach (two levels: pre and post) was included as a fixed factor. The lowest level of replication was an individual seine, $n = 162$ seines. As part of the PERMANOVA, we quantified variance components, the proportion of the multivariate variation accounted for by each variable (Searle et al. 1992; Anderson et al. 2008; Gerwing et al. 2016).

As a BACI design was used, two interaction terms are of particular interest (Underwood 1994, 1997). First, the Area $\times$ Breach interaction, if significant, would indicate that changes in the estuary beyond breaching may be influencing our findings. Second, the Site (Area) $\times$ Breach interaction elucidates which sites varied before and after breaching. Seven a priori planned contrasts were used to assess how individual sites varied before and after breaching. Of these a priori planned contrasts, the contribution of each fish species, including salmonids, to the observed change before and after breaching was explored using a SIMPER analysis (Clarke 1993; Clarke & Ainsworth 1993). As part of this analysis we calculated the ratio of each variable's average dissimilarity to the standard deviation of dissimilarities (Diss/SD). These values represent how consistently each variable contributed to observed differences; variables with a ratio greater than 1 consistently contributed to the difference, whereas those with a value below 1 did not (Gerwing et al. 2015). In this case a value over 1 would indicate that a given species is found consistently (in more seines than not) in higher or lower densities in a given location.

Goals 4 and 5: Homogenization of Water Chemistry in Channels and Pools. To elucidate if breaching homogenized water chemistry in pools and channels, PERMANOVAs were used; however, pool and channel chemistry was analyzed separately. With regards to pool water chemistry, as pH was only measured from the surface, pH was analyzed separately from other water chemistry variables (DO, salinity, and temperature). DO, salinity, and temperature data were normalized prior to analyses, and the response variable for this analysis was a resemblance matrix calculated using Euclidean distances. In the water chemistry PERMANOVA, site (two levels: Freshwater and Saltwater Pools), breach (two levels: pre and post), and depth (four levels: surface, 1, 2, and 3 m) were included as fixed factors. The base level of replication was an individual reading ($n = 42$). The PERMANOVA analyzing pH conditions in the pools was constructed as described above but excluded depth ($n = 12$). Water chemistry (DO, temperature, salinity, pH of surface water) of the channels was also analyzed using PERMANOVAs. Breach (two levels: pre and post) and location (two levels: inside and outside) were included as fixed factors, and the lowest level of replication was an individual reading ($n = 36$).

All analyses were performed in PRIMER with the PERMANOVA add-on (Anderson et al. 2008; Clarke & Gorley

2015), and water quality profiles were generated using R (Wickham 2016; R Core Team 2018). In all analyses, $\alpha = 0.05$ was used to indicate significance (Beninger et al. 2012).

Results

Goals 1–3: Enhancement of Salmonid Access and Monitoring of the Estuary Fish Community

The fish community varied significantly between sites, areas, before and after breaching, as well as over time (Table S1; Fig. S1). As the Area $\times$ Breach interaction was not significant ($p = 0.71$; Table S1), it is unlikely that changes in the estuary diminished our ability to detect impacts due to causeway breaching. Of the potential site contrasts within the Site (Area) $\times$ Breach interaction, only sites inside ($p = 0.04$) and outside ($p = 0.01$) the causeway, as well as South Beach ($p = 0.02$), varied significantly before and after causeway breaching (Table S1).

Fish communities inside and outside of the causeway varied by 41–44% before and after breaching (Table 1). The largest observed species difference both inside and outside the causeway (37 and 41% of the observed variation, respectively), and the only species that varied consistently before and after breaching was Three-spined Sticklebacks (*Gasterosteus aculeatus*). When contrasting populations before and after breaching, Three-spined Sticklebacks decreased inside the causeway, and increased outside the causeway (Table 1). Prickly Sculpins (*Cottus asper*) accounted for 18–22% of the observed variation, decreased both inside and outside the causeway, but did so in a non-consistent matter (Table 1). Chinook Salmon were not observed inside or outside the causeway prior to breaching but were present after. Chum Salmon were not observed outside of the causeway before breaching but were present post-breaching. Chum Salmon inside of the causeway decreased marginally before and after breaching. At the South Beach reference site, the fish community varied by 84% before and after breaching. Chum and Chinook Salmon decreased consistently after breaching, accounting for 42 and 25% of the variation, respectively (Table 1).

Variation before and after breaching with regards to fish communities in the Saltwater ($p = 0.64$) and Freshwater ($p = 0.33$) Pools was not statistically significant (Table S1). Despite this, important changes were observed in the fish communities (Tables 1 & S1). Of note, prior to breaching Chinook Salmon were absent in both pools; however, after breaching Chinook were observed in both pools. Chum Salmon were also absent in both pools prior to breaching, and after breaching Chum were observed in the Saltwater Pool. Finally, Coho Salmon were only observed in the Freshwater Pool prior to breaching, but following breaching Coho were observed in both pools.

Goals 4 and 5: Homogenization of Water Chemistry in Channels and Pools

Water quality in channels and pools varied spatially, as well as before and after breaching (Tables 2, S3 & S4;

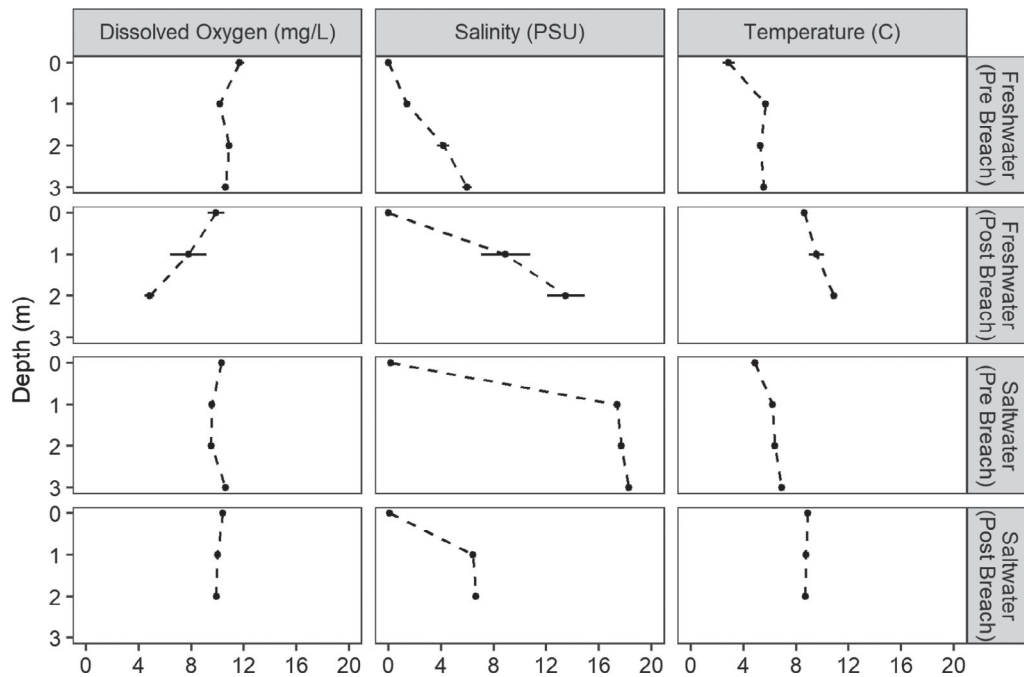
Table 1. Summary of similarities percentages (SIMPER) results investigating if the fish community of the Sarita Estuary varied before and after breaching of a causeway. Only a priori contrasts that were identified as statistically significant in the PERMANOVA (permutational multivariate analysis of variance, see Table S2) are presented. Species whose Diss/SD (dissimilarity over standard deviation of dissimilarity) was over 1, and therefore contributed consistently to observed differences, are indicated in bold text. Densities are fourth-root-transformed ($n = 162$).

Inside: Pre Versus Post							Outside: Pre Versus Post						
Average Dissimilarity = 41.66%							Average Dissimilarity = 44.34%						
Species	Pre-Average Density	Post-Average Density	Average Dissimilarity	Diss/SD	Contribution (%)	Cumulative (%)	Species	Pre-Average Density	Post-Average Density	Average Dissimilarity	Diss/SD	Contribution (%)	Cumulative (%)
<i>Gasterosteus aculeatus</i>	0.76	0.46	15.22	1.04	36.54	36.54	<i>Gasterosteus aculeatus</i>	0.20	0.53	18.07	1.31	40.74	40.74
<i>Cottus asper</i>	0.53	0.39	9.17	0.83	22.00	58.54	<i>Cottus asper</i>	0.56	0.50	8.16	0.94	18.39	59.14
<i>Syngnathus leptorhynchus</i>	0.18	0.03	6.54	0.95	15.70	74.24	<i>Oncorhynchus tshawytscha</i>	0	0.19	7.90	0.83	17.81	76.95
<i>Oncorhynchus tshawytscha</i>	0	0.12	3.99	0.69	9.58	83.81	<i>Syngnathus leptorhynchus</i>	0.05	0.09	4.51	0.68	10.17	87.12
<i>Oncorhynchus keta</i>	0.06	0.04	3.65	0.50	8.77	92.58	<i>Clinocottus acuticeps</i>	0.05	0.03	3.05	0.53	6.89	94.00
<i>Leptocottus armatus</i>	0.05	0	1.65	0.44	3.95	96.54	<i>Oncorhynchus keta</i>	0	0.05	1.97	0.39	4.44	98.44
<i>Clinocottus acuticeps</i>	0	0.03	1.04	0.32	2.49	99.02	<i>Cymatogaster aggregata</i>	0	0.02	0.69	0.22	1.56	100
SB: Pre Versus Post													
Average Dissimilarity = 83.97%													
Species	Pre-Average Density	Post-Average Density	Average Dissimilarity	Diss/SD	Contribution (%)	Cumulative (%)							
<i>Oncorhynchus keta</i>	0.67	0.05	34.86	2.10	41.51	41.51							
<i>Oncorhynchus tshawytscha</i>	0.44	0.26	20.84	1.27	24.82	66.33							
<i>Cottus asper</i>	0	0.21	12.73	0.75	15.16	81.49							
<i>Gasterosteus aculeatus</i>	0	0.19	10.13	0.81	12.06	93.56							
<i>Syngnathus leptorhynchus</i>	0	0.06	3.76	0.45	4.48	98.04							
<i>Porichthys notatus</i>	0	0.02	1	0.25	1.20	99.23							
<i>Leptocottus armatus</i>	0	0.02	0.64	0.26	0.77	100							

Table 2. Mean and standard error surface pH values of the Saltwater and Freshwater Pools (A) and water chemistry values in channels (B) before and after causeway breaching in the Sarita Estuary. *n* is the number of replicates.

(A)			
Site	Breach	n	pH
Freshwater Pool	Pre	12	8.34 ± 0.11
Freshwater Pool	Post	12	9.13 ± 0.29
Saltwater Pool	Pre	12	6.52 ± 0.02
Saltwater Pool	Post	12	7.98 ± 0.02

(B)						
Site	Breach	n	pH	DO (mg/L)	Temperature (°C)	Salinity (PSU)
Inside	Pre	9	7.59 ± 0.08	11.68 ± 0.09	5.59 ± 0.21	0.07 ± 0.01
Inside	Post	9	7.65 ± 0.01	10.44 ± 0.12	8.80 ± 0.00	3.27 ± 0.92
Outside	Pre	9	7.18 ± 0.05	11.49 ± 0.33	6.47 ± 0.18	11.19 ± 2.77
Outside	Post	9	7.67 ± 0.02	11.27 ± 0.22	8.82 ± 0.01	4.32 ± 0.99

**Figure 4.** Water quality profiles (mean and standard error) of the Saltwater and Freshwater Pools in the Sarita Estuary, before and after causeway breaching (*n* = 42).

Figs. 4, S2 & S3). In general, salinity in Freshwater and Saltwater Pools became more homogeneous post-breach compared to pre-breach (Table 2; Fig. 4). Water temperature consistently increased in both pools post-breach relative to pre-breach sampling (February). Regarding DO, concentrations remained consistent between the Freshwater and Saltwater Pools, pre- and post-breach, with the exception of the 2 m depth in the Freshwater Pool where mean DO decreased to 4.83 ± 0.38 mg/L (*n* = 3). Measurements at surface of pH in both the Freshwater and Saltwater Pools increased during the post-breach period (Table 2). In channels, salinity inside and outside of the causeway homogenized post-breach compared to pre-breach (Table 2), while water temperature consistently

increased at all channel sites post-breach. Both DO and pH remained consistent across channel sites pre- and post-breach (Table 2).

Discussion

In 2018, a causeway isolating the southern portion of the Sarita River estuary in Treaty Settlement Lands of the Huu-ay-aht First Nations on the west coast of Vancouver Island was breached to enhance salmon and tidal water access across the causeway. Immediately post-breach, fish community and water chemistry measurements indicated that these goals were achieved.

Goals 1–3: Enhancement of Salmonid Access and Monitoring of the Estuary Fish Community

Breaching of the Sarita causeway had an immediate impact upon the fish community, including salmonids, achieving the goals of improving access for salmon into both channel and pool habitat. Overall, the fish communities varied by 41–44% before and after breaching, with Three-spined Sticklebacks and Prickly Sculpins accounting for the majority of the observed variation. This is not surprising, as sticklebacks and sculpins were the most common fish in the estuary. While accounting for less of the overall variation, observed changes in salmon abundances with breaching were also observed. Chinook Salmon were not observed in channels inside or outside of the causeway before breaching but were observed post-breaching. Chum Salmon were not present outside of the causeway prior to breaching but were observed after breaching. Inside of the causeway Chum Salmon were present before and after breaching, and the post-breaching decrease was so marginal as to effectively indicate no change. These findings indicate that Chinook and Chum Salmon increased in abundance near the causeway (inside and outside), immediately making use of the breach and channel habitat on either side of the causeway. As such, the goal of breaching the causeway to promote juvenile salmon access through the causeway has been achieved. Previous alterations to causeways have exhibited similar impacts upon fish communities to those observed in this study (Eertman et al. 2002; Roni et al. 2002; AMEC 2015), particularly with juvenile salmon quickly making use of breaches and newly accessible habitat (Shreffler et al. 1990; Cordell et al. 1998; Simenstad & Cordell 2000; Gray et al. 2002).

Before breaching, the SB reference site had the highest observed densities of juvenile salmon in the Sarita Estuary. High densities at this location were surprising, as this exposed gravel beach is devoid of large woody debris, and sparse subtidal eelgrass beds offered little shelter for juvenile salmon from predators (Simenstad et al. 1982; Bugert & Bjornn 1991; Kennedy et al. 2018). Following breaching, salmon densities increased near the causeway, and decreased at SB. With only 1 year of data it is impossible to determine if this trend represents habitat selection. However, it is unlikely that migration out of the estuary by salmon was driving these trends, as the Area $\times$ Breach interaction was insignificant, and temporal differences beyond breaching of the causeway (Date) only accounted for 5.74% of the observed variation in the fish community. Interestingly, the opposite trend was observed with regards to Prickly Sculpins, which decreased inside and outside of the causeway following breaching, but increased at SB. Previous studies have observed high densities of Prickly Sculpins in areas dominated by gravel (White & Harvey 1999), and such a habitat preference may explain this observed spatial change. Continued monitoring is required to determine if these trends continue.

We observed no statistically significant differences before and after breaching with regards to the fish communities in the Freshwater and Saltwater Pools. Lack of statistical significance was a result of high variation between seines; however, important trends were identified pertaining to salmon. Specifically, prior to breaching Chinook and Chum Salmon were absent

from both pools, while post-breaching Chinook were observed in both pools and Chum were observed in the Saltwater Pool. Therefore, Chinook and Chum Salmon appear to have colonized these pools following breaching. Similarly, Coho Salmon were observed in the Freshwater Pool prior to breaching, and after breaching they were observed in both pools. Therefore, not only were Coho retained in the Freshwater Pool but breaching enabled Coho to become established in the Saltwater Pool as well. These pools represent potential high-quality rearing habitat for all three species of salmon given overhanging banks and vegetation, eelgrass beds, and depth of the pools (approximately 2 m at high tide), all of which provide shelter from predators (Simenstad et al. 1982; Quinn 2011; Carr-Harris et al. 2015). Therefore, continued use of these habitats could have positive impacts on salmon survival while in the estuary. Long-term monitoring is required to determine if salmon usage continues or expands; however, our goal of promoting Chinook and Chum access to these pools while retaining Coho habitat was successfully accomplished.

Assessments of non-salmonid fish species to assess overall breach success or failure suggested that several species require continued monitoring. One species requiring continual monitoring is Bay Pipefish (*Syngnathus leptorhynchus*), which is a common resident of eelgrass beds (Ryer & Orth 1987; Murphy et al. 2000). After breaching, Bay Pipefish decreased substantially inside of the causeway but increased slightly outside of the causeway and at SB. While it is likely that the observed decrease inside the causeway is a temporary artifact of Bay Pipefish reaching equilibrium in the newly connected habitat, more monitoring is needed to better assess this population's long-term trends. Further, we observed that low flows inside the causeway prior to breaching contributed to algal blooms that had a noticeable and negative impact upon eelgrass color and vigor. Such blooms have had devastating impacts upon other eelgrass beds in the past (Cosper et al. 1987; Nelson & Lee 2001). Following breaching, increased flow volume and velocities resulted in the removal of algal blooms and a rapid recovery of eelgrass stems. Given the new flow regimes, future monitoring will include qualitative assessments of eelgrass bed extent to determine if eelgrass beds expand. If this occurs, not only will it provide further habitat for juvenile salmon, but larger populations of Bay Pipefish may also be supported. Finally, sticklebacks have been shown to compete with juvenile salmonids and limit population growth in other restoration efforts (Gerwing & Plate 2018). As such, continued monitoring of stickleback populations is required to identify if expanding stickleback populations may be limiting salmonid population growth.

Goals 4 and 5: Homogenization of Water Chemistry in Channels and Pools

Breaching of the causeway had an immediate impact upon water chemistry, an impact that closely mirrors those of other causeway breaches, especially regarding enhanced tidal intrusion (McAlice & Jaeger 1983; Eertman et al. 2002; Gray et al. 2002; Roni et al. 2002; AMEC 2015). While there was still

variation in water chemistry following causeway breaching, in general re-connected pools and channels became more homogeneous. As such, our goals of homogenizing water chemistry in pools and channels were achieved. The greatest effect was observed in salinity, with the Saltwater Pool becoming less saline, and the Freshwater Pool becoming more saline after breaching. A decrease in DO from 9.88 ± 0.64 mg/L ($n = 3$) to 4.83 ± 0.38 mg/L ($n = 3$) was measured near the bottom of the Freshwater Pool post-breach. This measurement was below the instantaneous minimum DO water quality guideline for juvenile and adult salmon of 5 mg/L (ME 2018). The observed decrease in DO may be due to decreased solubility resulting from increases in both water temperature and salinity, but could also be attributed to increased biological reduction near the sediment–water interface (Kristensen 2000). Finally, channels inside of the causeway became more saline while channels outside became less saline. These observations not only show that the aquatic environment became more homogeneous after breaching of the causeway, but they also highlight the expansion of tidal waters into the southern portion of the estuary. Therefore, breaching was successful in increasing connectivity in this habitat.

Future Work

The long-term goal of causeway breaching was to enhance juvenile salmon access to high-quality rearing habitat in the Sarita Estuary to promote stabilization and enhancement of salmon stocks in the Sarita watershed. As such, water chemistry and the fish community will continue to be monitored for the next 4 years by the Huu-ay-aht First Nations. Such data will allow for a holistic assessment of breaching success, and elucidation of long-term response of salmon and other fish species to causeway breaching. Ongoing assessments of salmon escapement and fisheries in the Sarita watershed will help elucidate if causeway breaching has any long-term impacts upon salmon survival and abundance.

Regardless of the need for ongoing monitoring, breaching of a causeway in the southern portion of the Sarita Estuary had immediate impacts upon water chemistry and fish populations. These short-term indicators suggest that breaching was effective in mitigating fragmentation of the Sarita Estuary. As such, breaching improved access to 1.4 km² of habitat that had previously been inaccessible to juvenile salmon.

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Supporting Information

The following information may be found in the online version of this article:

Figure S1. Non-metric multidimensional scaling plot (nMDS) of the fish community.

Figure S2. Non-metric multidimensional scaling plot (nMDS) of water chemistry of the Saltwater and Freshwater Pools in the Sarita Estuary ($n = 42$).

Figure S3. Non-metric multidimensional scaling plot (nMDS) of surface water chemistry (dissolved oxygen [DO], temperature, pH and salinity) of channels before and after causeway breaching ($n = 36$).

Table S1. Mean and standard error of fish community densities (individuals/m²).

Table S2. Permutational multivariate analysis of variance (PERMANOVA) investigating if the fish community of the Sarita Estuary varied before and after breaching of a causeway.

Table S3. Permutational multivariate analysis of variance (PERMANOVA) investigating if water quality varied before and after breaching a causeway in the Freshwater and Saltwater Pools in the Sarita Estuary.

Table S4. Permutational multivariate analysis of variance (PERMANOVA) investigating if water quality varied before and after breaching a causeway of channels inside and outside of the causeway in the Sarita Estuary.

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